

Question ID: 5c94e6fa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$3x + 21 = 3x + k$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k ?

Correct Answer: 21

Rationale

The correct answer is **21**. It's given that the equation $3x + 21 = 3x + k$ has infinitely many solutions. If an equation in one variable has infinitely many solutions, then the equation is true for any value of the variable. Subtracting $3x$ from both sides of the given equation yields $k = 21$. Since this equation must be true for any value of x , the value of k is **21**.